

Nicholas Morse

Curriculum Vitae

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📍 Stockholm, Sweden
👤 Citizenship: USA

Researcher specializing in high-fidelity simulation (DNS/LES) of turbulent boundary layers, jets in crossflow, and multiphase flows, with expertise in turbulent heat and scalar transport, crossflow and curvature effects, and extreme-scale GPU computing.

Education

- 2023 **PhD**, *Aerospace Engineering & Mechanics*, University of Minnesota
Thesis: “High-fidelity unstructured overset simulation of complex turbulent flows” [↗](#)
Adviser: Professor Krishnan Mahesh
- 2020 **MS**, *Aerospace Engineering & Mechanics*, University of Minnesota
- 2018 **BAEM**, *Aerospace Engineering & Mechanics*, University of Minnesota
Minors: Math, Astrophysics

Experience

- May 2025 – Present **Postdoctoral Researcher**, *KTH Royal Institute of Technology*, Stockholm, Sweden
FLOW (Professors Philipp Schlatter, Ramis Örlü, and Mihai Mihaescu)
- Performed 18-billion-point DNS of a 3D turbulent boundary layer (TBL) using 3072 GH200 superchips on Europe’s first exascale supercomputer with the GPU-accelerated spectral element solver *Neko* [↗](#).
 - Established Reynolds-number scaling of wall heat transfer, turbulent stresses, and Reynolds analogy breakdown up to $Re_\tau \approx 2300$, the highest TBL friction Reynolds number simulated to-date.
 - Investigated streamline curvature and crossflow effects on turbulent kinetic energy and shear stress budgets of the Boeing bump turbulent boundary layer.
 - Developed a module in *Neko* for computation of turbulent heat/scalar flux and variance budgets.
- Jan – May 2025 **Group Leader: Multiscale CFD**, *Research Center Pharmaceutical Engineering*, Graz, Austria
- 2023 – 2025 **Senior Scientist**, *Research Center Pharmaceutical Engineering*, Graz, Austria
- Led the simulation strategy development for an EU Horizon 2020 project.
 - Coded a boundary element method from the ground up to resolve sub-Kolmogorov droplet breakup.
 - Unstructured curved elements, Gauss–Legendre quadrature, adaptive mesh refinement, feature-preserving mesh smoothing, adaptive time stepping, matrix-free GMRES(k), fast multipole method.
 - Implemented an ethanol-water mixture model in OpenFOAM to simulate impingement jet mixing.
- 2018 – 2023 **Graduate Research Assistant**, *University of Minnesota*, Minneapolis, MN, USA
Computational Fluids Lab (Professor Krishnan Mahesh)
- Performed large-scale (~ 10000 processor) simulations of complex turbulent flows using HPC.
 - Formulated streamline coordinates to study curvature in axisymmetric turbulent boundary layers.
 - Resolved experimental trip wires in LES to analyze model-scale boundary layer memory effects.
 - Identified mixing-enhancing secondary vortices from DNS and DMD of a jet in crossflow.
 - Extended a massively-parallel in-house unstructured finite-volume overset LES/DNS code.
 - Implemented *Hypre* GPU solvers, viscous flux corrections, and overset grid cutting algorithms.
- 2016 – 2017 **Undergraduate Research Assistant**, *University of Minnesota*, Minneapolis, MN, USA
Turbulent Shear Flow Lab (Professor Ellen Longmire)
- Designed tunable-buoyancy spheres to study particle transport in a turbulent boundary layer.
 - Manufactured and set up equipment for particle image velocimetry (PIV) experiments.
 - Measured the Stokes number dependence of particle restitution coefficients using high-speed imaging.

Skills

<i>Methods</i>	DNS, LES, spectral element method, unstructured finite volume, overset grids, boundary element method, fast multipole method, DMD, PIV	<i>Coursework</i>	Turbulence, compressible flow, gas dynamics, hypersonic flow, hydrodynamic stability, CFD (spectral, finite volume)
<i>Computing</i>	Fortran, MPI, Python, Matlab, Git; HPC: 3072 GPUs, 10,000 CPU cores	<i>CFD codes</i>	Neko, MPCUGLES, OpenFOAM, ANSYS, Basilisk
<i>Meshing</i>	Pointwise, GridPro, Gmsh, Salome	<i>Visualization</i>	ParaView, Tecplot, VisIt, PyVista
<i>Engineering</i>	SolidWorks, Simulink	<i>Manufacturing</i>	Machining, waterjet, composites

Publications

Journal articles

N. Morse, P. H. Alfredsson, M. Bode, N. Jansson, R. J. Lingwood, M. Mihaescu, R. Örlü, and P. Schlatter. Momentum and scalar transport in the von Kármán three-dimensional turbulent boundary layer. *Submitted to Journal of Fluid Mechanics*.

N. Morse. Insights into sub-Kolmogorov-scale droplet breakup via boundary element simulations [↗](#). *International Journal of Multiphase Flow*, 105460, 2025.

N. Morse and K. Mahesh. Tripping effects on model-scale studies of flow over the DARPA SUBOFF [↗](#). *Journal of Fluid Mechanics*, 975:A3, 2023.

N. Morse and K. Mahesh. Effect of tabs on the shear layer dynamics of a jet in cross-flow [↗](#). *Journal of Fluid Mechanics*, 958:A6, 2023.

N. Morse and K. Mahesh. Large-eddy simulation and streamline coordinate analysis of flow over an axisymmetric hull [↗](#). *Journal of Fluid Mechanics*, 926:A18, 2021.

Conference papers & abstracts

N. Morse, N. Jansson, M. Bode, R.J. Lingwood, P.H. Alfredsson, M. Mihaescu, R. Örlü, and P. Schlatter. Turbulence on a turntable: Extreme-scale DNS of the rotating-disk turbulent boundary layer [↗](#). *APS Gallery of Fluid Motion (Video)*, 2026.

N. Morse, P.H. Alfredsson, M. Bode, N. Jansson, R.J. Lingwood, M. Mihaescu, R. Örlü, and P. Schlatter. Revisiting von Kármán (1921): DNS of the rotating-disk turbulent boundary layer to $Re_\tau = 2300$. *79th Annual Meeting of the APS DFD*, 2026.

N. Morse, N. Jansson, M. Bode, R.J. Lingwood, P.H. Alfredsson, M. Mihaescu, R. Örlü, and P. Schlatter. Direct numerical simulation of the rotating disk turbulent boundary layer [↗](#). *14th International Symposium on Turbulence and Shear Flow Phenomena (TSFP14)*, 2026.

N. Morse, R. Stanly, R. Örlü, and P. Schlatter. Streamline curvature and crossflow effects on the turbulent boundary layer over a Gaussian bump [↗](#). *14th International Symposium on Turbulence and Shear Flow Phenomena (TSFP14)*, 2026.

N. Morse, M. Bode, N. Jansson, M. Mihaescu, R. Örlü, and P. Schlatter. Direct numerical simulation of the turbulent boundary layer on a rotating disk [↗](#). *15th ERCOFTAC Workshop on Direct and Large Eddy Simulation (DLES15)*, 2026.

N. Morse. Insights into sub-Kolmogorov droplet breakup via a boundary element method [↗](#). *BIRS Workshop: Particulates across Scales: Mathematical Modeling, Computation, and Applications*, 2025.

N. Morse, J. Remmelgas, and J. Khinast. A simulation framework for nanodroplet breakup in top-down nanoparticle production [↗](#). *AIChE Annual Meeting*, 2024.

M. Fenelon, Y. Zhang, L. Cattafesta, **N. Morse**, K. Mahesh, L. Li, and Z. Pan. Optimized timing schemes for multi-pulse shake-the-box particle tracking velocimetry [↗](#). *AIAA SciTech Forum*, 2023.

N. Morse and K. Mahesh. The shear layer structure of a tabbed jet in crossflow [↗](#). *75th Annual Meeting of the APS DFD*, 2022.

M. Fenelon, L. Cattafesta, Y. Zhang, K. Mahesh, and **N. Morse**. Optimizing dt for MP-STB in particle tracking velocimetry [↗](#). *75th Annual Meeting of the APS DFD*, 2022.

N. Morse, T. Kroll, and K. Mahesh. Large-eddy simulation of submerged marine vehicles [↗](#). *Proceedings of the 34th Symposium on Naval Hydrodynamics (SNH34)*, Washington, DC, 2022.

N. Morse and K. Mahesh. Streamline coordinate analysis of the flow past an axisymmetric body computed by large-eddy simulation [↗](#). *74th Annual Meeting of the APS DFD*, 2021.

N. Morse and K. Mahesh. Large-eddy simulation of appended submerged vehicles using an unstructured overset grid method [↗](#). *73rd Annual Meeting of the APS DFD*, 2020.

T. Kroll, **N. Morse**, W. Horne, and K. Mahesh. Large eddy simulation of marine flows over complex geometries using a massively parallel unstructured overset method [↗](#). *Proceedings of the 33rd Symposium on Naval Hydrodynamics (SNH33)*, Osaka, Japan, 2020.

N. Morse, W. Horne, and K. Mahesh. Towards large-eddy simulation of maneuvering vehicles using an unstructured overset grid method [↗](#). *72nd Annual Meeting of the APS DFD*, 2019.

D. Barros, Y. H. Tee, **N. Morse**, B. Hiltbrand, and E. Longmire. Investigation of particle lift off in a turbulent boundary layer [↗](#). *70th Annual Meeting of the APS DFD*, 2017.

Invited Talks

- 2026 *Revisiting von Kármán (1921): DNS of the rotating-disk turbulent boundary layer to $Re_\tau = 2300$* , ONERA, Toulouse, France
- 2023 *Exploring the physics of complex flows through high-fidelity unstructured overset simulations*, CSRI, Sandia National Laboratories, Albuquerque, NM, USA

Awards

- 2026 Invited Paper, Special Issue of *Flow, Turbulence and Combustion*
Best Video, Swedish E-Science Research Centre (SeRC) Annual Meeting
Erik Petersohns Memorial Foundation Grant
Semi-finalist, ERCOFTAC Milton Van Dyke Visualization Competition
- 2019 John A. & Jane Dunning Copper Fellowship for Aerospace Engineering & Mechanics
- 2018 Donald & Shirley Gorence Scholarship
- 2017 Robert H. & Marjorie F. Jewitt Fund Scholarship

Service

Co-organizer and session chair of a workshop at the *Banff International Research Station*: “Particulates across Scales: Mathematical Modeling, Computation, and Applications” [↗](#)
Reviewer for the *Journal of Fluid Mechanics*, *ASME Journal of Turbomachinery*, and *International Journal of Heat and Fluid Flow*.

Teaching & Mentorship

Developed and taught recitation exercises for a graduate-level turbulence course.
Led wind tunnel labs for laminar/turbulent boundary layer theory and self-similarity.
Mentored an undergraduate research assistant in overset DNS and simulation pre- and post-processing to simulate freely rising/falling particles in six-degree-of-freedom motion.